Smart Programming : YouTube Channel

An investment in Knowledge pays the best interest….

JDBC (PART 4)

Class.forName("----");

Class

=> class/interface : class (java.lang)

=> Defination : Class.forName("classname") - It simply loads a class into memory including its static initializers

=> Methods :

- forName("")
- getName()
- getPackage()
- getConstructors()
- getDeclaredClasses()
- getFields()
- getInterfaces()
- getMethods()
- ----- etc

Smart Programming
We Educate
We Develop

JDBC API (java.sql) :-

1. java.sql.Connection (I)
2. java.sql DriverManager (C)
3. java.sql.Statement (I)
4. java.sql.PreparedStatement (I)
5. java.sql.CallableStatement (I)
6. java.sql.ResultSet (I)

7. java.sql.DatabaseMetaData (I)
8. java.sql.ResultSetMetaData (I)

9. java.sql.Blob (I)
10. java.sql.Clob (I)

11. java.sql.Date (C)
12. java.sql.Time (C)
13. java.sql.TimeStamp (C)

Smart Programming
We Educate
We Develop

14. java.sql.Types (C)

---

1. java.sql.Connection (I)

=> class/interface : interface

=> Definition/Working :

- It is the session between java application & database
- The Connection interface is a factory of Statement, PreparedStatement, DatabaseMetaData etc i.e. it is used to get the objects for these interfaces (Statement, PreparedStatement, DatabaseMetaData)
- It provides methods for transaction management for eg commit(), rollback() etc

=> Methods :

- createStatement()
- createStatement(int resultSetType, int resultSetConcurrency)
- prepareStatement()

- setAutoCommit(boolean status)
- commit()
- rollback()
- close()

---

2. java.sql DriverManager (C)

=> class/interface : class

=> Definition/Working :

- It acts as an interface between java application and driver
- It keeps the track of drivers that are available and handles establishing a connection between database and suitable driver
- It mentains a list of Driver classes that have registered themselves by calling the method i.e. DriverManager.registerDriver()

=> Methods :

- registerDriver()
- deregisterDriver()
- getConnection(-,-,-)
- getDriver()

---

3. java.sql.Statement (I)

=> class/interface : interface

=> Definition/Working :

- It provides methods to execute the SQL queries with database
- It works as factory for ResultSet i.e. it provides factory methods to get the object of ResultSet

=> Methods:-

- executeQuery(-)
- executeUpdate(-)
- execute(-)

- executeBatch()

---

6. java.sql.ResultSet (I)

=> class/interface : interface

=> Definition/Working :

- It maintains a cursor pointing to a row of the table (which was retrieved from the database after firing sql query).
- The ResultSet cursor by default points before the first row

=> Methods :

- next()
- previous()
- first()
- last()
- absolute(-)
- relative(-)
- getString(-)

- getInt(-)

---

4. java.sql.PreparedStatement (I)

=> class/interface : interface

=> Definition/Working :

- It is sub interface of Statement interface (interface Statement interface PreparedStatement extends Statement)
- It is like Statement interface but its performance is much better as compared to Statement interface

=> Methods :

- executeQuery(-)
- executeUpdate(-)
- execute(-)
- executeBatch()
- setInt(-,-)
- setString(-,-)

5. java.sql.CallableStatement (I)

=> class/interface : interface

=> defination/working :

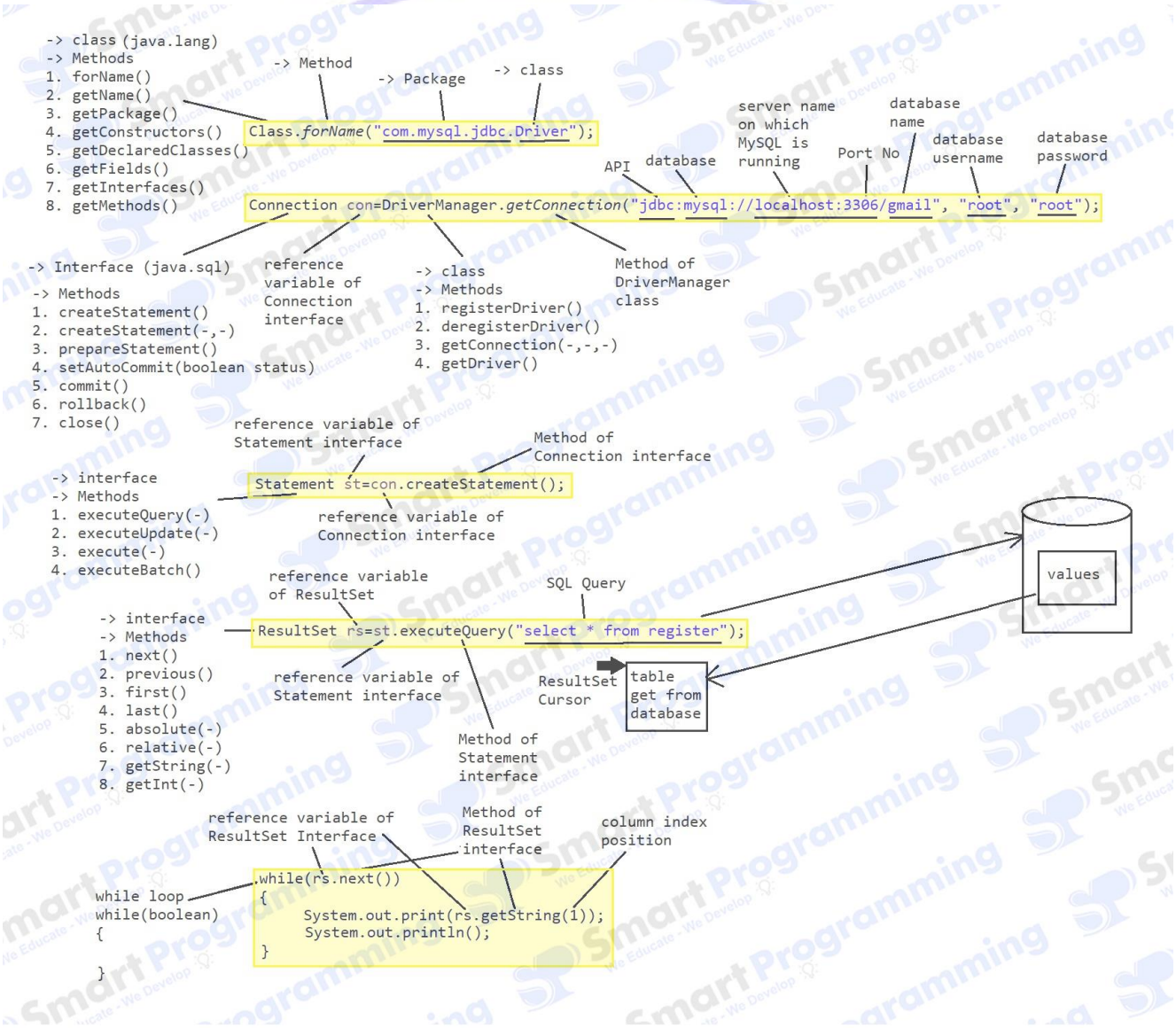
- It is used to call stored procedures and functions

=> Methods :

- executeQuery(-)
- executeUpdate(-)
- execute(-)
- executeBatch()
- setInt(-,-)
- setString(-,-)

Smart Programming
We Educate
We Develop

Diagram:-


-> class (java.lang)
-> Methods
1. forName()
2. getName()
3. getPackage()
4. getConstructors()
5. getDeclaredClasses()
6. getFields()
7. getInterfaces()
8. getMethods()
-> Method
-> Package
-> class
Class.forName("com.mysql.jdbc.Driver");
server name on which MySQL is running
database name
database username
database password
API
database
Port No
Connection con=DriverManager.getConnection("jdbc:mysql://localhost:3306/gmail", "root", "root");
-> Interface (java.sql)
-> Methods
1. createStatement()
2. createStatement(-,-)
3. prepareStatement()
4. setAutoCommit(boolean status)
5. commit()
6. rollback()
7. close()
reference variable of Connection interface
-> class
-> Methods
1. registerDriver()
2. deregisterDriver()
3. getConnection(-,-,-)
4. getDriver()
Method of DriverManager class
reference variable of Statement interface
Method of Connection interface
Statement st=con.createStatement();
-> interface
-> Methods
1. executeQuery(-)
2. executeUpdate(-)
3. execute(-)
4. executeBatch()
reference variable of Connection interface
reference variable of ResultSet
SQL Query
values
-> interface
-> Methods
1. next()
2. previous()
3. first()
4. last()
5. absolute(-)
6. relative(-)
7. getString(-)
8. getInt(-)
ResultSet rs=st.executeQuery("select * from register");
reference variable of Statement interface
ResultSet Cursor
table get from database
Method of Statement interface
reference variable of ResultSet Interface
Method of ResultSet interface
column index position
while loop
while(boolean)
{
}
while(rs.next())
{
System.out.print(rs.getString(1));
System.out.println();
}

Company Links & Contacts

Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

Websites: https://www.smartprogramming.in/

https://courses.smartprogramming.in

Android App:
https://play.google.com/store/apps/details?id=com.smartprogramming

YouTube Channel:
https://www.youtube.com/c/SmartProgramming